Orchestration for the Edge–Cloud Continuum

What is the Edge–Cloud Continuum?

- **Cloud:** centralized, high-performance infrastructure (data centers)
- **Edge:** computing resources closer to users/devices
- **Continuum:** seamless collaboration between cloud and edge resources
- **Goal:** deploy, manage, and scale workloads dynamically across both environments

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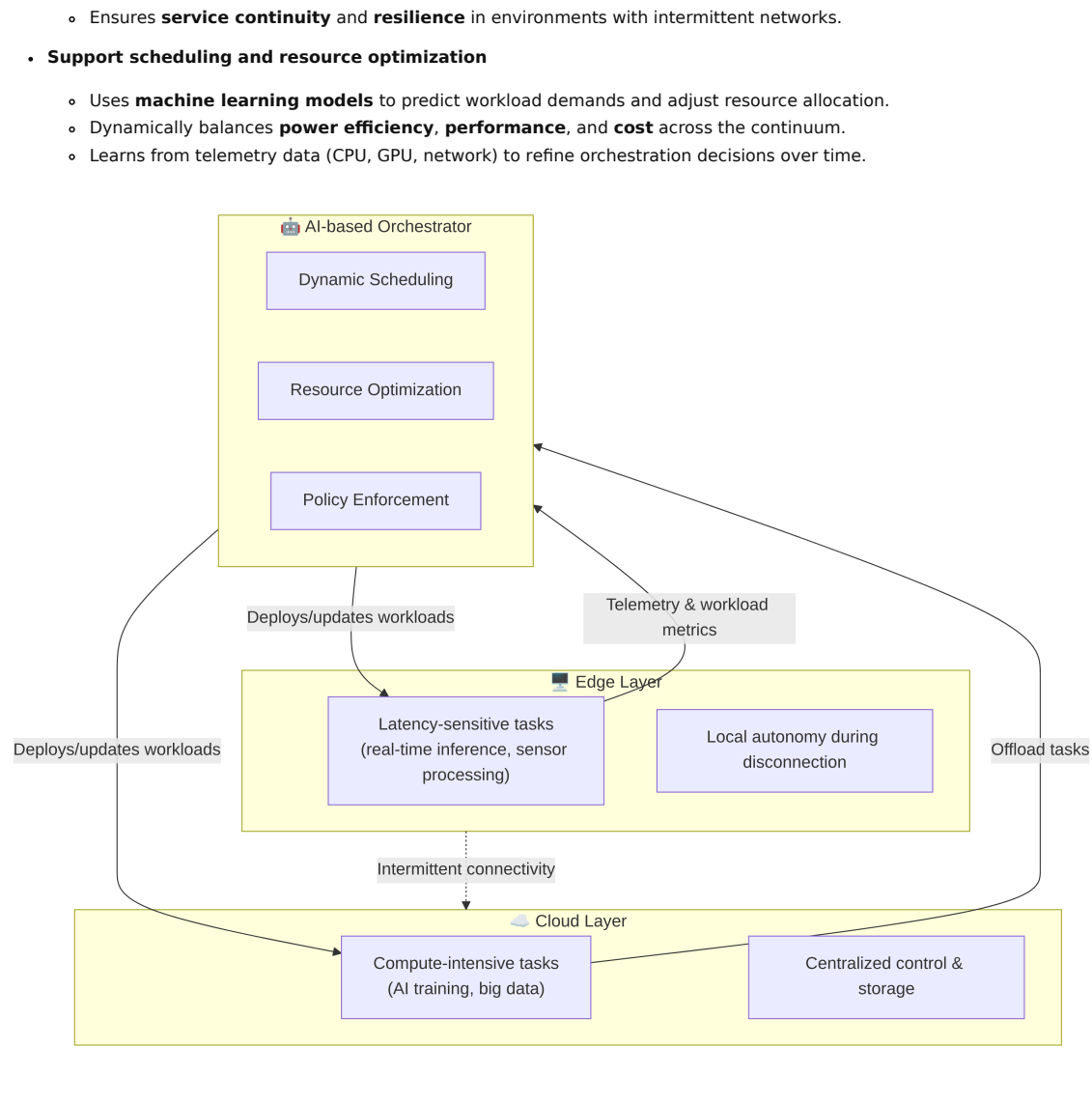
Why Orchestration?

- Edge and cloud environments are **heterogeneous**
- Applications are **distributed**, often involving:
  - IoT devices
  - Real-time analytics
  - AI inference
- Need for **automation** in:
  - Deployment
  - Scaling
  - Resilience
  - Monitoring

Toward Unified and Adaptive Orchestration

- **Manages both latency-sensitive and compute-intensive tasks**
  - Deploys **real-time workloads** (e.g., sensor fusion, anomaly detection) close to data sources at the edge.
  - Offloads **compute-heavy operations** (e.g., model training, batch analytics) to the cloud.
  - Continuously adapts based on **network latency**, **resource availability**, and **QoS requirements**.
- **Enables autonomous operation during disconnection**
  - Edge nodes maintain **local decision-making** even when cloud connectivity is lost.
  - Synchronization and reconciliation occur automatically when the connection is restored.
  - Ensures **service continuity** and **resilience** in environments with intermittent networks.
- **Support scheduling and resource optimization**
  - Uses **machine learning models** to predict workload demands and adjust resource allocation.
  - Dynamically balances **power efficiency**, **performance**, and **cost** across the continuum.
  - Learns from telemetry data (CPU, GPU, network) to refine orchestration decisions over time.


🤖 AI-based Orchestrator
Dynamic Scheduling
Resource Optimization
Policy Enforcement
Deploys/updates workloads
Telemetry & workload metrics
🖥️ Edge Layer
Deploys/updates workloads
Latency-sensitive tasks (real-time inference, sensor processing)
Local autonomy during disconnection
Offload tasks
Intermittent connectivity
☁️ Cloud Layer
Compute-intensive tasks (AI training, big data)
Centralized control & storage

Kubernetes (K8s)

- **De facto standard** for container orchestration in the cloud
- Provides:
  - Automated deployment and scaling
  - Service discovery and load balancing
  - Self-healing (restarts, rescheduling)
- Suitable for **large-scale** clusters
- Challenges at the edge:
  - High overhead
  - Limited connectivity and resources

KubeEdge: Extending K8s to the Edge

- Open-source project from **CNCF**
- Extends Kubernetes **beyond the cloud** to edge nodes
- Architecture:
  - **CloudCore:** runs in the cloud (control plane)
  - **EdgeCore:** runs on edge devices (data plane)
- Features:
  - Offline tolerance
  - Device management APIs
  - Synchronization between edge and cloud
  - Works with AI and IoT workloads

K3s: Lightweight Kubernetes

- **Optimized for edge and IoT** environments
- Developed by Rancher (now SUSE)
- Key features:
  - Single binary (<100 MB)
  - Minimal dependencies
  - SQLite instead of etcd (by default)
  - ARM-compatible
- Ideal for **resource-constrained devices**

MicroK8s: Developer-Friendly Kubernetes

- Developed by **Canonical** (Ubuntu)
- **Zero-ops** Kubernetes for:
  - Local development
  - Edge and IoT
- Key features:
  - Easy installation: `snap install microk8s --classic`
  - Modular via `microk8s enable <addon>`
  - Supports clustering (HA setups)
- Runs on Linux, macOS, and Windows

References

- CNCF KubeEdge: https://kubeedge.io
- SUSE K3s: https://k3s.io
- Canonical MicroK8s: https://microk8s.io
- Kubernetes Docs: https://kubernetes.io